ECON 101: Macroeconomics
Solutions – Quiz 4

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Part A: Multiple Choice Solutions

1. **Correct answer: (C)**
   A **unit of account** means money is used to quote prices and record values. Listing a car price as $28,000 uses money as a common measure of value.

2. **Correct answer: (C)**
   The simple deposit multiplier is:

$$\text{Multiplier} = \frac{1}{r_d} = \frac{1}{0.20} = 5$$

3. **Correct answer: (B)**
   An open-market **sale** removes reserves from the banking system. As reserves fall, banks have fewer funds to support deposits and loans, so the money supply tends to fall.

4. **Correct answer: (C)**
   Currency held by the public is part of the narrow money supply because it is highly liquid and can be used directly for transactions.

5. **Correct answer: (C)**
   If banks hold excess reserves, they lend out less than the maximum possible amount. As a result, the actual increase in the money supply is **smaller** than the amount predicted by the simple multiplier.

Part B: Long Numerical Problem Solution

6. Money Creation and the Banking System

Given:

- Initial deposit = \$8,000
- Desired reserve ratio $r_d = 0.10$
- Banks lend out all excess reserves
- The public redeposits all loan proceeds

(a) Desired reserves held by Bank A

Desired reserves are calculated as:

$$\text{Desired reserves} = r_d \times \text{deposit}$$

Substitute the values:

$$\text{Desired reserves} = 0.10 \times 8{,}000 = 800$$

$$\boxed{\text{Desired reserves} = \$800}$$

(b) Amount Bank A can initially lend out

The bank can lend out its **excess reserves**, which equal:

$$\text{Excess reserves} = \text{deposit} - \text{desired reserves}$$

Substitute:

$$\text{Excess reserves} = 8{,}000 - 800 = 7{,}200$$

$$\boxed{\text{Initial loan} = \$7{,}200}$$

(c) Simple deposit multiplier

The simple deposit multiplier is:

$$m = \frac{1}{r_d}$$

Substitute:

$$m = \frac{1}{0.10} = 10$$

$$\boxed{m = 10}$$

(d) Maximum total increase in deposits

The maximum total increase in deposits is:

$$\Delta D = m \times \text{initial deposit}$$

Substitute:

$$\Delta D = 10 \times 8{,}000 = 80{,}000$$

$$\boxed{\text{Maximum total increase in deposits} = \$80{,}000}$$

Interpretation: Starting from the initial \$8,000 deposit, repeated lending and redepositing can eventually generate total deposits of \$80,000 across the banking system.

(e) Maximum total increase in loans

There are two equivalent ways to calculate this.

Method 1: Total loans = Total deposits − Total reserves

We know total deposits eventually become:

$$80{,}000$$

Since the reserve ratio is 10%, total desired reserves must be:

$$0.10 \times 80{,}000 = 8{,}000$$

So total loans are:

$$80{,}000 - 8{,}000 = 72{,}000$$

$$\boxed{\text{Maximum total increase in loans} = \$72{,}000}$$

Method 2: Use the first-round logic

The first bank lends \$7,200. This amount is multiplied through the banking system:

$$7{,}200 + 6{,}480 + 5{,}832 + \cdots$$

This geometric series sums to:

$$72{,}000$$

So:

$$\boxed{\text{Maximum total increase in loans} = \$72{,}000}$$

(f) Why the actual increase in deposits may be smaller

The answer in part (d) assumes:

- banks lend out *all* excess reserves,
- households redeposit *all* borrowed funds,
- there are no leakages from the system.

If these assumptions do not hold, the actual increase in deposits will be smaller.

Reason 1: Banks hold excess reserves
If banks choose to keep some reserves instead of lending them out, then less money is loaned and the deposit-creation process becomes weaker.

Reason 2: Households hold cash
If households keep part of the loan proceeds as currency rather than redepositing them, then less money returns to the banking system. This reduces further rounds of lending.

Conclusion:

$$\boxed{\text{The actual increase in deposits will be smaller than } \$80{,}000}$$

because both excess reserves and cash holdings reduce the multiplier process.

End of Solution Key